

① Show $m(a+bX) = a + b \cdot m(X)$

$$m(X) = \frac{1}{N} \sum_{i=1}^N X_i$$

$$m(a+bX) = \frac{1}{N} \sum_{i=1}^N a + bX_i$$

$$= \frac{1}{N} \left(\sum_{i=1}^N a + \sum_{i=1}^N bX_i \right)$$

$$= \frac{1}{N} \left(\sum_{i=1}^N a + b \sum_{i=1}^N X_i \right)$$

$$= \frac{1}{N} \left(aN + b \sum_{i=1}^N X_i \right)$$

$$= a + \frac{b}{N} \sum_{i=1}^N X_i$$

$$= a + b \cdot m(X)$$

Thus it is Proven $m(a+bX) = a + b \cdot m(X)$

② Show $\text{Cov}(X, a+bY) = b \cdot \text{Cov}(X, Y)$ $\text{Cov}(X, a+bY) = \text{Cov}(X, Y)$

$$\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y))$$

$$\begin{aligned}\text{Cov}(X, a+bY) &= \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(a+bY_i - m(a+bY)) \\ &= a+bY_i - (a+b \cdot m(Y)) \\ &= \cancel{a} + bY_i - \cancel{a} - b \cdot m(Y) \\ &= bY_i - b \cdot m(Y) \\ &= b(Y_i - m(Y))\end{aligned}$$

$$= \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) \cdot b(Y_i - m(Y))$$

$$= \frac{b}{N} \sum_{i=1}^N (x_i - m(X))(Y_i - m(Y)) = b \cdot \text{Cov}(X, Y)$$

Thus it is Proven,
 $\text{Cov}(X, a+bY) = b \cdot \text{Cov}(X, Y)$

$$\begin{aligned}
 \text{Cov}(a+bX, a+bX) &= \frac{1}{N} \sum_{i=1}^N (a+bX_i - m(a+bX)) (a+bX_i - m(a+bX)) \\
 \textcircled{3} \quad &= \frac{1}{N} \sum_{i=1}^N (a+bX_i - (a+b \cdot m(X))) (a+bX_i - (a+b \cdot m(X))) \\
 &= \frac{1}{N} \sum_{i=1}^N (\cancel{a}+bX_i - \cancel{a} - b \cdot m(X)) (\cancel{a}+bX_i - \cancel{a} - b \cdot m(X)) \\
 &= \frac{1}{N} \sum_{i=1}^N (bX_i - b \cdot m(X)) (bX_i - b \cdot m(X)) \\
 &= \frac{1}{N} \sum_{i=1}^N b(X_i - m(X)) \cdot b(X_i - m(X)) \\
 &= \frac{b^2}{N} \sum_{i=1}^N (X_i - m(X)) (X_i - m(X)) = b^2 \cdot \text{Cov}(X, X)
 \end{aligned}$$

$$\begin{aligned}
 \text{Cov}(X, X) &= \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (X_i - m(X)) \\
 &= \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2 = S^2 = \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2
 \end{aligned}$$

Thus it is proven $\text{Cov}(a+bX, a+bX) = b^2 \text{Cov}(X, X)$
 and $\text{Cov}(X, X) = S^2$

4. No, transforming the mean is different than transforming the values.

$$M(X) = \frac{1+2+3}{3} = 2 \quad g(M(X)) = 4$$

$$g(x) = x^2$$

$$M(g(x)) = \frac{1^2+2^2+3^2}{3} = \frac{14}{3} \quad \frac{14}{3} \neq 4$$